Manit Khera

Third-year Software Engineering (Honours) student at Deakin University, applying for the Early Career Data Scientist role at Syncrowin

0412 761 206
manitkhera26@gmail.com
github.com/uiqvb
linkedin.com/in/manit-khera-603957252

HOW THE PROJECTS CONNECT `Arduino + six sensors` → `Python on a Raspberry Pi` → `my .NET backend` → `5,000 simulated robots on Azure`

About me

Hi, I'm Manit. Most of what I've built in the last year has been around one small robot. I made it for my embedded systems unit, connected it to a backend I wrote for a different unit, and then used that backend in a third unit to see how it coped with thousands of simulated robots.

Your application asks about a data science project I'm proud of. I should be upfront: I haven't trained a model, and I haven't studied statistics, probability, linear algebra or machine learning yet. The closest thing I have is the work I did with this robot's sensor data, so that's what most of this portfolio is about.

The part I enjoyed most across all three projects was figuring out why something wasn't behaving the way I expected, usually by going back through the data. I'd like to do a lot more of that, on real industrial data, with people who know the maths better than I do.

About the robot

BODY	Two wheels, six sensors, and a cardboard shell
USED FOR	Moving washing, my Open Day demo, and as the model for 5,000 simulated copies
GOOD AT	Getting the washing out of the rain
BAD AT	Wonky tape, and low batteries
MARKS	93 and 100/100

Why Syncrowin

I read through your site before applying, and what stuck with me was that your recommendations come with the reasoning behind them instead of just an answer. I ended up building my robot's backend in a similar way, for a less noble reason: I couldn't trust what the robot told me, so the server checks every report before saving it. I'd like to work on that kind of problem with data from real sites, in a small team where I can learn from the engineers directly.

Smart Clothesline Robot

Embedded Systems unit · 93

Arduino C++ · Raspberry Pi · Python/Flask · MQTT · Demo video · Write-up · Code

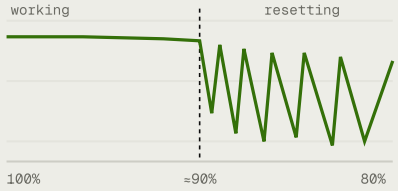
THE PROBLEM The robot moves washing between a sunny spot and a sheltered spot by following a tape line. An Arduino reads six sensors and sends the readings every five seconds over MQTT to a Python program I wrote on a Raspberry Pi, which decides what to do.

WHAT I DID The rain sensor was cheap and not very reliable, so the robot also checks the weather forecast before it moves. I tested this by wetting the sensor, and by feeding it fake rain readings to make sure it would go with the forecast.

The harder problem was power. On fresh batteries everything worked, but once they dropped to around 90 percent the robot started resetting at random. I assumed the batteries were the problem and kept buying new ones, probably about $100 worth, before I went back through the logs. The boot ID and reset cause in each message showed the board was browning out, which led me to a faulty voltage converter.

WHAT I LEARNED I should have looked at the logs much earlier. I now add fields like those to everything I build, because they are what made the fault findable. I also still have a pile of batteries sitting at about 90 percent.

ROBOT BEHAVIOUR VS BATTERY CHARGE



A rough sketch from memory, not recorded data.

Robot Controller API

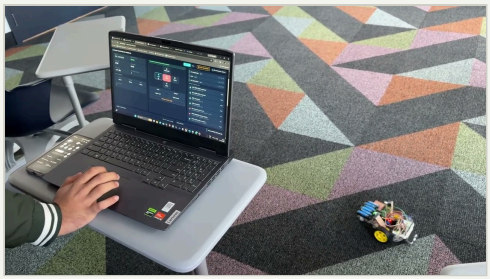
Backend Development unit · 100/100 · shown at Deakin Open Day

.NET 8 · C# · PostgreSQL · Docker · Jenkins Open Day demo Code

THE PROBLEM For my backend unit I wrote an API that does the robot's thinking. The robot just asks the server what to do next. When I connected it to the clothesline robot, the server and the robot often disagreed about where the robot actually was.

WHAT I DID I started logging every command and result to a history table in PostgreSQL, so I could see what had really happened. I also planned for the Wi-Fi dropping out, even though it never did. Each command comes with instructions for undoing it, so if the robot hears nothing for seven seconds it backs up along its last moves and then reports what it did. The server runs that report through a simulation before saving it.

HOW IT WENT When I tested it on the real robot, it stopped exactly where the server predicted. The project got 100/100 and was picked for Deakin Open Day, where I demoed it to prospective students and their parents.



My dashboard driving the robot (bottom right).

~120
API endpoints

7 s
without a reply before it backs up

Scaling a Robot Dispatch API

Distributed Programming unit

C# · ASP.NET Core · PostgreSQL · Azure · NBomber · xUnit Demo video Code Report

THE PROBLEM For a third unit I wanted to see what would happen with far more robots. I simulated a fleet on Azure and ran it against my backend. At 5,000 robots it didn't finish a single job.

WHAT I DID I went through the code and found 12 problems, including a race where two robots could claim the same job. I made job claims atomic in PostgreSQL, added a limit so the server tells robots to retry instead of falling over, and added a second server. Then I ran 80 benchmark runs on Azure, repeating each grid setup three times and taking the median so one lucky run couldn't skew the result.

WHAT I LEARNED At 400 robots the median time to claim a job went from 766.5 ms to 59.5 ms, with 54 automated tests behind the changes. I never found the new limit, because my load generator ran out of capacity before the backend did. I wrote the method and results up in a report.

JOBS FINISHED PER SECOND, 400 ROBOTS

Before	177
1 server	608
2 servers	1,069

JOBS FINISHED PER SECOND, 5,000 ROBOTS

Before	0
2 servers	~1,008

From my benchmark results on Azure. Both charts use the same scale.

What I haven't done yet

Statistics, probability, linear algebra and machine learning. I haven't built a forecasting model or used PyTorch. So far my work has been collecting sensor data and checking whether it can be trusted, not modelling it.

What I'd like to learn at Syncrowin

The modelling side, properly, from people who do it every day. I learn fastest when there's a real problem in front of me, and each project here started as a uni assignment that I kept adding to.

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